CI/CD Pipeline for a Student Attendance Management Portal

Build · Test · Gate · Containerise · Deploy · Provision

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The problem

- Attendance written on paper and keyed in later
- A student learns their standing weeks afterwards
- Eligibility disputes have no audit trail

9–14 days

measured delay from mark to student visibility

A system of record

- Faculty opens a session; students mark themselves
- Closing records everyone who did not as absent
- Only extreme cases escalate to a conversation

Immediate

the student sees the record as it is written

The pipeline — eleven stages, one verdict

1	2	3	4	5	6	7	8	9	10	11
Checkout	Build	Unit test	Package	Integration	Quality gate	Build image	Deploy container	Deploy Tomcat	Verify	Post

The gate sits at stage 6, before the image is built. If a browser journey fails, no image is published and nothing deploys. A check that runs after deployment can only report what users already have. The gate tests a throwaway container started from the very artefact that ships if it passes.

What each tool is for

Docker
Packages the app with its runtime, and provides the lab: eight containers standing in for machines.

Jenkins
Decides when a change ships. Eleven stages, defined in a Jenkinsfile versioned with the code.

Selenium
Drives a real browser through real tasks, so the gate tests the page a human sees.

Ansible
Declares the state a host should be in and converges it. Agentless, over SSH.

Puppet
The same job, masterless, in its own declarative language. Both are demonstrated.

Maven
One command compiles, tests and packages, so the pipeline needs no bespoke build steps.

Verification

103 unit — the business rules

7 integration — web layer and persistence

19 browser — a real browser, real tasks

67.9% line coverage overall, **85.8%** across the service layer where the business rules live. The 19 browser journeys contribute nothing to that figure: they drive a separate JVM the coverage agent never sees.

What the tests caught

- **The roster rendered blank.** A template read a property that no longer existed, so the page silently hid its own Stop and Strike buttons. It passed every unit test.
- **A promise the code did not keep.** The page said an override was logged against your name; only the role was recorded.
- **An escalated case vanished** from the dashboard count.

None would fail a compile. None would be obvious in a demonstration.

Results

129	11	16	< 2 min	0 days
tests, zero failing	pipeline stages, all green	versioned images in the registry	to record a session (SC1)	until a student sees it (SC2)

One artefact is promoted, never rebuilt: the image tagged **1.0.19-6be7e00** — build number plus commit — is the one that reaches staging and production alike. Never only **:latest**, because a tag that moves cannot be rolled back to.

What this project does not solve

No database migrations
Schema managed by ddl-auto with no Flyway. A NOT NULL column cannot be added to a populated table — in production, data loss.

Observability stops at a health check
Up or down. Nothing reports slow, or failing for a fraction of users.

Single node
No load balancing, no blue-green, and a window during deployment where the service is down.

DevOps is replacing "I think it works" with evidence — and automating the step so the evidence appears whether or not anyone asks

github.com/Swaroop192005/CI-CD-Pipeline-for-a-Student-Attendance-Management